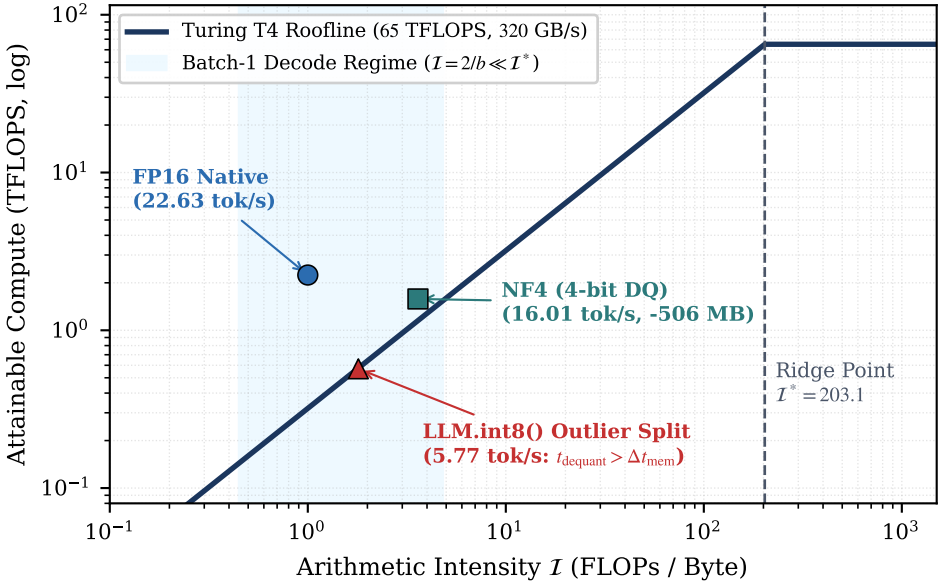
(a) Proposition 1: Batch-1 Decoding Roofline & Dequantization Bound



(b) Propositions 2 & 3: Subspace Dichotomy & Spectral Rank Truncation

Attention Routing Subspace $\mathcal{M}_{\text{attn}}$

$\{W_q, W_v\}$ (qv) $\subset \{W_q, W_k, W_v, W_o\}$ (qkvo)
Saturates Syntactic Schemas (JSONOut: 100.0%)

Spectral Rank Filter ($r \le r_{\text{eff}} \ll d_{\text{model}}$)

$\Delta W = (\alpha/r)BA$, $r^* \in [4, 16]$ prevents overfitting
26M LoRA (45.71%) > 26M Full SFT (36.80%)

FFN Key-Value Memory $\mathcal{M}_{\text{ffn}}$

SwiGLU $\{W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}\}$
Required for Domain QA (+13.36% on CourseQA)

DPO Step-Size Trust Region ($\eta \le 5 \times 10^{-6}$)

$\nabla_\theta \mathcal{L}_{\text{DPO}} \propto \sigma(-u)(\nabla \log \pi_w - \nabla \log \pi_l)$
Prevents Off-Manifold Likelihood Displacement